

Energy Audit Report - Victoria Island Cluster

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Approved by: Chukwuemeka Eze

Audit Date: April 10, 2026

1. Audit Scope

This audit covered TWR-VI-002 (Victoria Island) and TWR-VI-005 (Eko Hotel), with a full energy consumption assessment of installed equipment and operational practices.

3. Total Daily Consumption

- TWR-VI-002: 18.4 kWh
- TWR-VI-005: 14.2 kWh

5. Power Factor Reading

- TWR-VI-002: PF = 0.84
- TWR-VI-005: PF = 0.91

6. Energy Waste Identified

- TWR-VI-002: Old-style fluorescent lighting still installed, consuming 0.6 kW unnecessarily.
- TWR-VI-002: Air conditioner set at 18°C, adjusted to 24°C during audit to reduce excessive cooling load.

8. Grid vs Generator Cost Comparison

- Grid Supply Cost per kWh: NGN 85
- Generator Supply Cost per kWh: NGN 220

9. Recommendations and Priority

1. Immediate: Replace fluorescent lighting with LED fixtures at TWR-VI-002.
2. Immediate: Maintain AC setpoint at 24°C to reduce cooling load.
3. Short-term: Install capacitor bank at TWR-VI-002 to improve power factor.
4. Ongoing: Monitor grid quality and maintain generator service intervals to reduce reliance on generator supply.

10. Sign-off

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Date: April 10, 2026

2. Equipment Inventory

Equipment	Rated Power (kW)	Measured Power (kW)	Running Hours/Day	Daily kWh
BTS/RRU	2.5	2.3	24	55.2
Transmission Equip.	1.8	1.6	24	38.4
Rectifier	1.2	1.1	24	26.4
Cooling (AC units)	3.5	3.2	12	38.4
Lighting	0.8	0.7	10	7.0
Security System	0.5	0.4	24	9.6
Battery Chargers	1.0	0.9	8	7.2

4. Grid Supply Quality

Time of Day	Voltage (V)	Frequency (Hz)	THD %
Morning	228	50.1	3.2
Afternoon	231	49.9	3.5
Evening	226	50.0	3.1

7. Savings Opportunity

Item	Current Cost	Recommended Change	Estimated Annual Saving
Fluorescent Lighting	NGN 420,000	Replace with LED fixtures	NGN 180,000
Cooling Setpoint	NGN 1,200,000	Adjust to 24C	NGN 350,000
Power Factor (VI-002)	NGN 980,000	Install capacitor bank	NGN 220,000